

8/09/22

TASK - 5

JOIN QUIRES, EQUIVALENT AND RECURSIVE QUIRES

AIM:- TO implement and execute Joins,
equivalent queries and Recursive queries in SQL

PROCEDURE:

1. Create Table DEPARTMENT & STUDENT
2. Insert the values into tables
3. Perform Join operation
4. Perform equivalent & recursive query
5. Display result

```
CREATE TABLE DEPARTMENT (
    DEPT ID INT PRIMARY KEY,
    DEPT NAME VARCHAR(50);
```

```
CREATE TABLE STUDENT (
    STU - ID INT PRIMARY KEY,
    AGE INT,
    DEPT ID INT,
    FOREIGN (DEPT ID)
    REFERENCES DEPARTMENT (DEPT ID)
);
```

```
INSERT INTO DEPARTMENT VALUES
(201, 'Computer science');
(202, 'Electronics');
(203, 'Mechanical');
```


INSERT INTO STUDENT VALUES

- 1) ('Ravi', 20, 201),
- 2) ('Sneha', 22, 201),
- 3) ('Amit', 19, 202),
- 4) ('Priya', 24, 203),
- 5) ('Kiran', 23, 201);

SELECT * FROM DEPARTMENT 4;

	DEPTID	DEPTNAME
1	201	Computer Science
2	202	Electronics
3	203	Mechanical

SELECT * FROM STUDENT 4;

	STU-ID	NAME	AGE	DEPTID
1	1	Ravi	20	201
2	2	Sneha	22	201
3	3	Amit	19	202
4	4	Priya	24	203
5	5	Kiran	23	201

SELECT S.NAME S.AGE ID-DEPTNAME
FROM STUDENT S
INNER JOIN DEPARTMENT D
ON S.DEPTID = D.DEPTID;
-- INNER JOIN.

	NAME	AGE	DEPTNAME
1	Ravi	20	Computer Science
2	Sneha	22	Computer Science
3	Amit	19	Electronics
4	Priya	24	Mechanical
5	Kiran	23	Computer Science

-- LEFT OUTER JOIN
 SELECT S.NAME, S.AGE, D.DEPTNAME
 FROM STUDENT US
 LEFT JOIN DEPARTMENT D
 ON S DEPT ID = D DEPTID;

	NAME	AGE	DEPT NAME
1	Ravi	20	COMPUTER SCIENCE
2	Shrha	22	COMPUTER SCIENCE
3	Amit	19	ELECTRONICS
4	Priya	24	MECHANIS
5	Hiron	23	COMPUTER SCIENCE

SELECT S.NAME, S.AGE, D.DEPARTNAME
 FROM STUDENT S
 RIGHT JOIN DEPARTMENT D
 ON S-DEPT ID = D-DEPT ID;

	NAME	AGE	DEPTNAME
1	Ravi	20	COMPUTER SCIENCE
2	Shrha	22	COMPUTER SCIENCE
3	Hiron	23	COMPUTER SCIENCE
4	Amit	19	ELECTRONICS
5	Priya	24	MECHANICAL

SELECT TOP 3 S.NAME, S.AGE, D.DEPTNAME
 FROM STUDENT US
 FULL OUTER JOIN DEPARTMENT UP
 ON S-DEPT ID = D DEPT.ID;

	NAME	AGE	DEPT NAME
1	Ravi	20	COMPUTER SCIENCE
2	Shrha	22	COMPUTER SCIENCE
3	Amit	19	ELECTRONICS

--- EQUIVALENT QUERIES

--- USING JOIN

SELECT S.NAME, S.AGE

FROM STUDENT S

JOIN DEPARTMENT D ON S.DEPTID = D.DEPTID

WHERE D.DEPTNAME = ('COMPUTER SCIENCE')

	NAME	AGE
1	Ravi	20
2	Sheha	22
3	Kiran	23

-- RECURSIVE QUERIES

WITH COUNT CTE AS

SELECT 1 AS N

UNION ALL

SELECT N+1

FROM COUNT CTE

WHERE N < 5

)

SELECT * FROM COUNT CTE;

	N
1	1
2	2
3	3
4	4
5	5

VELTECH	
EX No.	
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	6
MIXED VOICE (5)	5
RECORD (5)	4
TOTAL (20)	20
SIGN WITH DATE	2/8/22

RESULT: Thus, Implementation of Join queries, Equivalent and Recursive queries has successfully executed and verified.